Tim Jacobs

Senior/Staff Software Engineer | Data, AI & Platform Architecture

CO 80501, Longmont, Colorado (US)

https://www.linkedin.com/in/tim-jacobs-software-engineer timj.cu@gmail.com (720) 934-6361

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SUMMARY

Staff Software Engineer with deep expertise in large-scale data platforms, LLM-powered retrieval systems (RAG), and cloud-native infrastructure. Proven architect of event-driven pipelines, vector search applications, and shared platform components that power data-intensive products. 15+ years building backend systems from storage validation to distributed AI services.

SKILLS

Data Engineering, AI & Platform Architecture

Expert

Snowflake | dbt | Apache Airflow | RAG architecture | Vector search & embeddings | Event-driven systems

AI & Machine Learning

Accomplished

Snowflake Cortex | LLM integration & prompting | Streamlit | Custom tokenization & vectorization | k-NN / semantic search

Cloud Architecture & DevOps

Expert

AWS | Kubernetes | Terraform | Docker | CI/CD pipeline design | Microservices architecture

Software Development

Expert

Python | SQL | Go/Golang | REST API | FastAPI

Databases & Storage

Expert

PostgreSQL | MySQL | MongoDB | Redis | DynamoDB | Elasticsearch | Database design & optimization

Engineering Leadership

Expert

Technical architecture & roadmap | Team mentorship & growth

WORK EXPERIENCE (4)

Dec 2022 - Current

Senior Software Engineer at EDO Inc.

Data platform & AI systems architecture for cross-platform content intelligence.

- Architected an LLM mapping pipeline on Snowflake Cortex (RAG, custom tokenization/vectorization, k-NN similarity search); reviewers accepted the LLM pick for 85% of 17,000+ mapped keys in its first six months.
- Built the Streamlit review app that replaced the PR-based seed-file workflow, raising throughput 4.2x (5,151 vs 1,220 decisions/week) and cutting engineer PR reviews from 511 in 18 months to zero; 7 reviewers now write directly.
- Implemented medallion architecture on Snowflake-based data pipelines to ingest and process trillions of rows of cross-platform data, supporting 20+ enterprise partners.
- Engineered a hypernormalized metadata management system that standardized fragmented data from dozens of sources into a single canonical schema, achieving 80% coverage of all monthly events.

- Shipped 120+ work items across two delivery spaces in 12 months, including 12 highest-priority (P0) items and the Phase 1 integration of a major media company's advertising data platform.

Oct 2020 - Dec 2022

Staff Software Engineer at Pie Insurance

Full-stack platform engineering, quality-focused technical leadership, and operational data architecture.

- Architected and deployed 40+ shared platform components consumed by multiple engineering teams; reduced time-to-market for new features by 50% and improved code reuse across the organization.
- Led technical design of CI/CD pipeline modernization; improved deployment velocity by 3x, reduced mean time to recovery by 60%, and enabled 200+ deployments per week.
- Led operational data team; designed and implemented reference data re-architecture using event-driven ETL; reduced mean commit-to-deploy time from weeks to hours, increasing release velocity by 3x.
- Developed scalable test automation framework for microservices in Python/Golang using AWS Lambda, S3, and SQS; automated 90% of previously manual test execution, reducing test cycle time from 6 hours to 12 minutes.
- Mentored 6 junior engineers on microservices patterns, testing strategy, and AWS cloud best practices; 3 engineers promoted during tenure.

Apr 2016 - Oct 2020

Senior Software Engineer at NetApp - Solidfire

Platform architecture, DevOps transformation, and test infrastructure leadership.

- Designed and led a test result reporting platform from concept to production; engineered a queryable database with 2M+ daily test records and automated management dashboards, eliminating 40+ hours per week of manual reporting.
- Spearheaded a DevOps transformation initiative with cross-functional stakeholders; reduced build failure rate by 50%, improved deployment reliability by 35%, and decreased infrastructure costs by $500K+ per year.
- Architected a containerized test framework enabling migration of 1,000+ test cases from legacy systems to modern CI-driven infrastructure; reduced test execution time by 70% and improved reproducibility.

Jun 2013 - Apr 2016

SSD Validation Engineer → Engineering Supervisor at Micron Technology

Validation leadership, data-driven operations, and team management (4 direct reports).

- Managed a team of 4 engineers validating enterprise/data center SAS/SATA SSDs across 4 Tier-1 OEM partners; maintained >$60M quarterly revenue through rigorous, on-time qualification processes.
- Designed and deployed a qualification test status database with automated test script integration; eliminated manual reporting, enabled real-time visibility into validation progress, and reduced management reporting overhead by 30 hours per week.

EDUCATION (1)

- 2005

Bachelor of Science Electrical Engineering at University of Colorado at Boulder

Grade: 3.74 (Cum Laude)